

Education

LAHORE UNIVERSITY OF MANAGEMENT SCIENCES

Lahore, Pakistan

Aug 2022 - Jun 2026

- BSc in Computer Science with a specialization in Artificial Intelligence.
- First Year Dean's Honor List
- Relevant Coursework: Multi-Modal Diffusion Models, AI on Edge, Large Language Model Systems
- Study Abroad: **Gwangju Institute of Science & Technology** - Gwangju, South Korea · Aug 2024 - Dec 2024
Relevant Coursework: Advanced Deep Learning, Machine Learning, AI Experience Lab (NLP), Data Science in Healthcare

Experience

AI Research Engineer I (promoted from AI Intern) | Skylabs AI

July 2025 - March 2026 - Present

- Improving the performance of open-source VLMs (hosted on a multi-GPU setup) on a SOTA visual-reasoning benchmark.
- Created & operated autonomous and human-in-the-loop agentic pipelines for VLM finetuning & synthetic data creation.

Research Intern | Large Language Model Compression | LUMS

Jan 2025 - June 2025

- Performed a literature review on compression and mechanistic interpretability. Conducted research on pruning-based LLM compression, focusing on Wanda.
- Helped develop a contrastive learning based compression framework that works with existing iterative pruning architectures, such as magnitude pruning and Wanda, and gives a better accuracy with higher sparsity ratios.

Research Intern | Machine Learning and Computer Vision Lab | GIST

Aug 2024 - Dec 2024

- Developed and evaluated four image captioning architectures - LSTM, LSTM with skip connections, spatial attention, and multi-head self-attention - on the Flickr8k dataset using BLEU-1 to BLEU-4 metrics, with the multi-head self-attention model achieving the best performance.

Publication

Mechanistic Evidence for Preserved-but-Misaligned Representations in Non-IID FedAvg

June 2026

Accepted at the ICML 2026 Workshop on Mechanistic Interpretability (Co-author)

- Applied mechanistic interpretability techniques, including sparse circuit extraction, linear probing, Universal Sparse Autoencoders, and SVCCA, to analyze Federated Learning under severe non-IID data distributions.
- Demonstrated that FedAvg largely preserves learned representations despite severe accuracy degradation, identifying representation misalignment as a complementary explanation to client drift and objective inconsistency.

Projects

Mechanistic Analysis of Object Hallucination in Vision-Language Models

- Investigated the impact of ZCA-whitened CLIP features on object hallucination in connector-only Vision Language Models using Q-Former and MLP connectors, evaluating multiple whitening strengths and architectures.
- Used linear probes, CKA, attention diagnostics, and decision-margin analysis to show that whitening improves hallucination performance for Q-Former models but degrades MLPs, identifying readout alignment as the primary mechanism.

Hessian-Based 4-bit Post-Training Quantization for Diffusion Models | 10xEngineers Collab

- Implemented Qronos post-training quantization using second-order Hessian (H) and correction (G) matrices, performing FP64 curvature accumulation and error-aware block-wise transformer reconstruction for stable low-bit compression
- Reduced model size (1.16GB → 1.04GB) with near-FP16 semantic alignment (CLIP 32.63 vs 32.65) and competitive perceptual fidelity (LPIPS 0.20, FID 0.61).

Agentic Framework for Business Data Analysis

- Built an end-to-end agentic framework using fine-tuned Mistral 7B and Gemini APIs to automate exploratory data analysis for non-technical business users.
- Implemented an agentic RAG framework using Flitx and LangChain that sources through 800+ PDFs for Pakistan and industry-specific insights for business queries.

Technical Skills

Languages: Python, C++, C, Java, Haskell, TypeScript, MATLAB

Libraries: PyTorch, TensorFlow, Keras, Scikit-learn, NumPy, Pandas, Matplotlib, Seaborn, TVM

Specialized Frameworks: Google ADK, Git, Hugging Face (Transformers, Datasets), Lifelines (Survival Analysis), XGBoost, Jupyter